ANUPAM VISHWAKARMA

AI/ML Engineer | Generative AI | Python Developer

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ABOUT ME

Results-driven AI/ML Engineer & Python Developer with 7+ years of experience designing and deploying enterprise-scale Generative AI and machine learning solutions across publishing, energy, cybersecurity, and fintech domains. Currently at Springer Nature Technology & Publishing Solutions, Pune, building production grade Gen AI systems including Alternative Text Generation, Taxonomy Classification, Ethics Compliance Validation, and Budget Optimization platforms leveraging LLMs (OpenAI, Gemini), RAG pipelines, LangChain, vector databases, and prompt engineering. Proficient in Python, Django, FastAPI, Google Cloud Platform, Docker, PostgreSQL, and MongoDB, with hands-on expertise in REST APIs, distributed systems (Celery & Redis), and AI observability using Langfuse, Grafana, and Prometheus. Holds an M. Tech in Information Technology from IIIT Allahabad, B. Tech in Computer Science, and a Professional Certificate in Business Management from IIM Kozhikode, with Generative AI and Machine Learning certifications from Coursera.

CORE COMPETENCIES

▸ **Generative AI & LLM Engineering** Designing and deploying production LLM pipelines using OpenAI, Gemini, and open-source foundation models with prompt engineering and fine-tuning strategies.	▸ **RAG & Knowledge Systems** Building Retrieval Augmented Generation (RAG) architectures with LangChain, vector databases (Chroma DB, Pinecone), semantic search, and embeddings for grounded AI responses.
▸ **Machine Learning & Deep Learning** End-to-end ML model development from feature engineering and model training to evaluation, optimization, and production deployment using Scikit-learn, TensorFlow, and PyTorch.	▸ **Computer Vision & OCR** Developing vision-language pipelines using YOLO v7, TrOCR, and multimodal LLMs for object detection, image classification, and intelligent document/meter reading.
▸ **NLP & Text Processing** Applying NLP techniques including named entity recognition, text classification, semantic similarity, topic modeling, and information extraction for domain-specific corpus.	▸ **MLOps & AI Observability** Implementing monitoring, logging, and performance evaluation pipelines using Langfuse, MLflow, Grafana, and Prometheus for cost optimization and model analytics in production.
▸ **Full-Stack AI Application Dev** Building end-to-end AI-powered web apps using Django, FastAPI, Streamlit, and GCP with secure authentication, scalable APIs, and intuitive user interfaces.	▸ **Data Engineering & ETL** Designing large-scale ETL workflows for data extraction, transformation, and loading across structured (PostgreSQL, MySQL) and unstructured (XML, JSON, CSV) data sources.
▸ **Cloud & DevOps** Deploying scalable microservices on Google Cloud Platform using Docker, Jenkins, and CI/CD pipelines, managing distributed task queues with Celery and Redis.	▸ **Google Workspace & Marketplace** Developing and publishing enterprise Google Workspace Add-ons (Docs/Sheets) using Google Apps Script, successfully deployed to Google Marketplace for organizational use.

KEY SKILLS

- **Languages:** Python, Java, C/C++, JavaScript, HTML5/CSS, Google Apps Script
- **GenAI & LLMs:** OpenAI, Gemini, Claude, Prompt Engineering, Fine-Tuning
- **RAG & Vector DBs:** LangChain, Chroma DB, Pinecone, FAISS, Semantic Search, Embeddings, Document Retrieval
- **ML / DL Frameworks:** Scikit-learn, TensorFlow, PyTorch, YOLO v7, TrOCR, Hugging Face Transformers
- **NLP:** Text Classification, Summarization, Named Entity Recognition, spaCy, NLTK, Topic Modeling
- **Web Frameworks:** Django, Django REST Framework, FastAPI, Streamlit
- **Cloud & DevOps:** Google Cloud Platform (GCP), Docker, Virtual Machines,
- **Databases:** PostgreSQL, MySQL, MongoDB, SQL Server, Redis
- **Task Queues:** Celery, Redis
- **AI Observability:** Langfuse, MLflow, Grafana, Prometheus, Model Monitoring, Cost Optimization
- **Testing:** Pytest, Sonar Cube, Unit Testing, Integration Testing
- **Agile / Tools:** JIRA, Confluence, Git/GitHub, Agile, Scrum, Kanban

PROFESSIONAL EXPERIENCE

Springer Nature Technology & Publishing Solutions, Pune

AI/ML Engineer | April 2023 – Present | Team Size: 12 | Stack: Python, Django, FastAPI, LangChain, OpenAI, Gemini, GCP, Streamlit, Apps Script, JIRA

Project 1: Budget & Campaign Optimization Platform

Tech Stack: *Python, Django, Streamlit, ML Model Fine-Tuning, GCP, REST APIs*

Summary: AI-driven marketing analytics platform enabling data-driven budget allocation and campaign strategy testing across multi-regional markets. Delivers revenue impact projections through fine-tuned ML models and interactive visualization dashboards.

Key Contributions: Built Django-based UI/UX with fine-tuned predictive models, developed region-wise continental visualization dashboards, deployed scalable REST APIs on GCP for global marketing teams.

Project 2: Augmented Insights: LLM-Powered Content Creation Tool

Tech Stack: *Google Apps Script, Google Workspace APIs, LLMs, Google Marketplace*

Summary: Human-in-the-loop Gen AI platform enabling editorial augmentation of academic content using large language models. Deployed as a production Google Workspace Add-on and published on Google Marketplace.

Key Contributions: Architected and deployed Google Docs Add-on via Apps Script integrated with LLMs, published on Google Marketplace, implemented real-time LLM content suggestions with editorial control workflows.

Project 3: EBM Database: Data Engineering & SAP Integration

Tech Stack: *Python, XML Parsing, ETL Pipelines, SAP, CSV*

Summary: Large-scale data engineering pipeline for extracting, cleaning, and profiling editorial board member (EBM) data from Jflow XML feeds across multiple journals, enabling targeted scholarly communication.

Key Contributions: Built ETL workflows for XML-to-CSV transformation, performed data deduplication, profiling, and merging, automated SAP server uploads for downstream CRM and communication workflows.

Project 4: LLM Vision Model Benchmarking

Tech Stack: *Python, FastAPI, Streamlit, Vision-Language Models, PDF Processing, Cost Analysis*

Summary: Research and development initiative to evaluate cost-effective alternatives to GPT-4 Vision for academic image understanding. Included automated PDF-to-image extraction, multi model benchmarking, and cost performance analysis.

Key Contributions: Developed PDF extraction pipeline by ISBN, implemented multi-model vision inference via FastAPI, benchmarked open-source VLMs against GPT-4V on accuracy and cost, deployed server-side inference with Streamlit frontend.

Project 5: Alt Text Generator: Vision Language AI for Academic Images

Tech Stack: *Python, GenAI, Vision-Language Models (VLMs), FastAPI, Streamlit, Langfuse, Grafana, Prometheus*

Summary: Production Gen AI system generating accessible alternative (alt) text for scientific figures, charts, and diagrams in academic journals and books using multimodal vision-language models improving accessibility and SEO compliance.

Key Contributions: Designed end-to-end VLM inference pipeline, integrated Langfuse for prompt observability and cost tracking, deployed Grafana + Prometheus monitoring, built REST API backend (FastAPI) with Streamlit user interface.

Project 6: Taxonomy Finder: Intelligent Content Classification Engine

Tech Stack: *Python, GenAI, LLMs, LangChain, NLP, Vector Databases*

Summary: Automated content classification system using Generative AI and ML to map structured and unstructured academic content into predefined hierarchical taxonomies with subcategory.

Key Contributions: Designed LLM powered classification pipeline using LangChain and semantic embeddings, built hierarchical taxonomy mapping logic, evaluated multi-model accuracy for category, subcategory.

Project 7: Ethics Compliance Validator: AI Powered Regulatory Review

Tech Stack: *Python, GenAI, LLMs, NLP, Document Analysis, FastAPI*

Summary: Gen AI compliance automation system to evaluate research manuscripts for ethical adherence verifying IRB approvals, informed consent, animal study protocols, and regulatory disclosure statements.

Key Contributions: Built document ingestion and NLP analysis pipeline, applied LLM reasoning for ethics clause detection and validation, generated structured compliance reports with flagged gaps and recommendations.

[MDP INFRA India Pvt. Ltd (Client: MPMKVVCL MP State Power Co.), Bhopal]

Team Lead / Python Developer / ML Engineer | Oct 2021 – March 2023 | Team: 24 | Stack: Python, Django, DRF, ML, YOLO v7, TrOCR, PostgreSQL, GIT

Project 1: QCPortal: Vendor & NABL Lab Onboarding Platform

Tech Stack: *Django, Django REST Framework, PostgreSQL, Bootstrap, HTML/CSS/JS*

Summary: Enterprise grade web portal serving 50,000+ users for registration and onboarding of Vendors, NABL accredited labs, Turnkey Contractors, and MPCZ officers. Includes Central Purchase System for vendor payment management.

Key Contributions: Developed Django backend with PostgreSQL, built REST APIs for user management and document workflows, implemented role-based access control, automated email triggers and document tracing. Live at: qcportal.mpcz.in

Project 2: Electricity Theft Detection: ML Based Anomaly System

Tech Stack: *Python, Scikit-learn, Pandas, NBS Billing Data, ML Classification Models*

Summary: Machine learning-powered fraud detection system identifying abnormal electricity consumption patterns and potential theft from NBS billing data reducing revenue losses and improving operational efficiency for MPCZ.

Key Contributions: Designed feature engineering pipelines on historical consumption data, trained and evaluated classification models for theft detection, built analytics dashboards to surface suspicious usage patterns.

Project 3: Intelligent Meter Reading: YOLO v7 + TrOCR Pipeline

Tech Stack: *Python, YOLO v7, TrOCR (Vision-Language Model), FastAPI, OCR, Computer Vision*

Summary: End-to-end automated meter reading pipeline using YOLO v7 for display region detection and TrOCR (Transformer-based OCR) for digit extraction directly from meter images eliminating manual reading errors.

Key Contributions: Designed two stage Computer Vision pipeline (detection -> OCR), deployed inference via REST APIs, built data extraction workflows for downstream billing integration, achieved high-accuracy automated digit recognition.

CyberInfrastructure Pvt. Ltd, Indore

Software Developer (ML & Python) | March 2021 – Oct 2021 | Team: 5 | Stack: Python, Streamlit, FastAPI, Django, ML Algorithms

Project 1: Risk Maturity Assessment Platform

Tech Stack: *Python, Streamlit, Django, FastAPI, ML, Secure Auth*

Summary: AI/ML-powered enterprise risk assessment web platform enabling corporate users to complete structured questionnaires and receive automated Risk Maturity scores with analytical reports.

Key Contributions: Built ML scoring engine and FastAPI backend, developed interactive Streamlit dashboards, implemented secure JWT authentication and user management, automated PDF report generation.

Project 2: Defense ML Module: Israel Military Force Client

Tech Stack: *Python, Machine Learning, Modular Code Architecture*

Summary: Developed secure, modular Python code modules leveraging classical ML algorithms for a US and Israeli defense client outsourced through CyberInfrastructure.

Key Contributions: Architected reusable Python ML modules, applied classical ML algorithms for classification and pattern recognition, maintained strict code modularity and documentation standards for defense-grade delivery.

Aspirify Enterprise Pvt. Ltd, New Delhi

Python Programmer (AI/ML) | Aug 2020 – March 2021 | Team: 6 | Stack: Python, Django, Celery, Redis, ML, Pandas

Project 1: AI Powered Network Vulnerability Assessment System

Tech Stack: *Django, Celery, Redis, Python, ML Pipelines, Security Analytics*

Summary: Distributed, real-time network scanning and vulnerability assessment platform using AI/ML for automated risk detection, threat scoring, and security insight generation across enterprise networks. (Defense & R&D domain)

Key Contributions: Built Django web app with Celery-Redis distributed task queues for parallel network scans, integrated ML based anomaly detection, developed security analytics dashboards and automated threat reporting.

Innoviti Payment Solutions Pvt. Ltd, Bangalore

Data Science & ML Engineer Intern | Feb 2020 – May 2020 | Stack: Python, Pandas, NumPy, Matplotlib, PostgreSQL, Jupyter

Conducted end-to-end data analysis, ML model development, and feature engineering on multi-bank payment and transaction datasets. Wrote optimized PostgreSQL queries in Jupyter Notebooks for data extraction, built predictive ML models and BI dashboards for transaction pattern analysis and business intelligence.

SNSE, New Delhi

Python Developer | July 2017 – Aug 2019 | Team: 4 | Stack: Python, Django, ML, OpenCV, NLP, REST APIs, MySQL, AWS, Web Scraping, ROS

Project 1: Knowledge Management & Intelligent Q&A Portal

Tech Stack: *Python, Django, REST APIs, NLP, MySQL, HTML, Bootstrap*

Summary: Domain-specific knowledge aggregation and Q&A platform with NLP-powered content search, categorization, and retrieval supporting user-contributed knowledge repositories.

Key Contributions: Built Django REST APIs, implemented NLP-based text processing for content organization, designed database schemas with optimized search queries, developed responsive frontend with Bootstrap.

Project 2: AI-Based Victim Detection for Rescue Robots

Tech Stack: *Python, OpenCV, Computer Vision, Face Detection, ROS*

Summary: Real-time computer vision system for autonomous rescue robots detecting human victims in disaster environments through face detection and image analysis algorithms integrated with ROS.

Key Contributions: Built CV pipelines using Python and OpenCV, implemented face detection and object localization, integrated vision modules with ROS robot framework, optimized model accuracy for low light/occluded environments.

Project 3: Recommendation Engine: Collaborative Filtering & Content Based ML

Tech Stack: *Python, Scikit-learn, Pandas, NumPy, ML, EDA*

Summary: Personalized recommendation system using collaborative filtering and content-based ML models to generate product/content recommendations based on user preference and item similarity patterns.

Key Contributions: Built and evaluated collaborative filtering models, performed EDA and feature engineering, implemented similarity metrics and ranking algorithms, generated business intelligence insights from predictive analytics.

Project 4: Python Automation, ETL & Microservices Development

Tech Stack: *Python, REST APIs, Web Scraping, ETL, Django, MySQL, AWS*

Summary: Delivered multiple client projects spanning data analytics, large scale web scraping, ETL pipeline development, and RESTful microservice APIs for business intelligence and process automation.

Key Contributions: Designed ETL workflows for structured/unstructured data ingestion, built web scraping solutions, developed RESTful microservice APIs, implemented data visualization and reporting for client BI requirements.

EDUCATION

- **M. Tech in Information Technology** IIIT Allahabad (Indian Institute of Information Technology), Uttar Pradesh [2019]
  Research: DNA & Genome Sequencing, Olfaction Detection via ML, Pattern Matching
- **Professional Certificate Program in Business Management (PCPBM)** IIM Kozhikode (Indian Institute of Management), Kerala [2022]

CERTIFICATIONS

- Generative AI Leader Google Cloud Certificate [2026]
- Generative AI for Everyone Coursera [2024]
- Machine Learning Specialization Coursera (Andrew Ng) [2023]
- Chatbot Builder Training Yellow.ai Academy [2022]
- Data Science Web App with Streamlit Coursera [2020]

- Python for Data Science   Intellipaat [2020]
- Machine Learning Training   Internshala [2019]
- Advance Java Certification   C-DAC Jaipur [2015]

ACHIEVEMENTS & PUBLICATIONS

- IEEE Publication: "Predicting Smell Perception from Molecular Descriptors using Machine Learning" IEEE En&T 2020, Dolgoprudny, Russia. ISBN: 978-1-7281-8830-0, Page 331.
- Paper: "Mobile Virtual Class" Imperial Journal of Interdisciplinary Research (IJIR), 2017, Vol-3, Issue-4.
- Best Paper Award: "iDoc" National Conference on Smart Computation and Technology (NCSCT), Poornima Institute, Jaipur, April 2016. Published in IJCTT V35(5).
- Paper: "Brain Chip as the Solution for the Development of Society" NCRIE, Apex Group of Colleges, Jaipur, April 2014. Published in IJEMS.
- Organizer Certificate: International Conference on Bioinformatics and System Biology 2018, IIIT Allahabad.

Declaration: I hereby declare that all the above-mentioned particulars are true to the best of my knowledge and belief.

Place: Pune, Maharashtra, India | Date: 10 July 2026   *(Anupam Vishwakarma)*